Binary numbers

Counting and adding!

What is a number?

Decimal number – also known as “base 10”:

5,407

What is a number?

Decimal number:

5407 =
5 thousand 4 hundred and 7

What is a number?

Columns matter – each digit has inherent value (“five”) and positional value (“thousands”)

Thousands	hundreds	tens	ones
5	4	0	7

Can we generalise?

10^3	10^2	10^1	10^0
Thousands	hundreds	tens	ones
5	4	0	7

Can we generalise... more?

For a base n (in this case n = 10)

n^3	n^2	n^1	n^0
Thousands	hundreds	tens	ones
5	4	0	7

General positional rule

For any base n the columns encode values of rising powers.

n^x	...	n^2	n^1	n^0

What values?

In decimal (base 10, remember), each column can contain digits with a value, and they have the values 0-9.

Can we generalise again?

Base 10 digits go from 0-9 – there are ten actual digits (ten values), and the highest value is 10 – 1.

So for any base n there are n actual digits, and the highest value is n - 1

Wasn't this about binary?

Now we can just apply those general rules to the specific case of binary.

Can we generalise... less?

For a base n (in this case n = 2)

n^3	n^2	n^1	n^0
2^3	2^2	2^1	2^0
eights	fours	twos	ones

Can we generalise less again?

For any base n there are n actual digits, and the highest value is n – 1

So “base 2” digits go from 0 - 1 – there are 2 actual digits (2 values), and the highest value is 2 – 1 (or just 1, if you like)

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Counting

The only interesting bit about counting is when to “overflow” or “carry” (we use those terms a lot)

When we get to the highest value, which in base ten is “9”, then we have “9 ones”, and the next number isn't “10 ones” – we can't put “10” in the ones column because 9 is our highest value. So it's “1 lot of 10”

Counting in binary

Same thing, but the highest value comes up a lot quicker!

1 is the highest value, So after 1, we don't have “2 lots of 1” – there's no 2 digit. So we have “1 lot of 2” because there's a 2s **column.**

Counting in binary

1 plus 1 is 1 lot of 2, and therefore no more ones:

2^1	2^0
1	0

2 in binary is “10”. To distinguish that from “10” in base ten (one lot of ten....), we sometimes use 0b as a prefix. 10 is ten, 0b10 is 2. Usually it will be clear from context.

Counting in binary

Now we have an “empty” ones column, so 2 + 1 is “1 lot of 2 and 1 lot of 1”

2^1	2^0
1	1

3 = 0b11

Counting in binary

Then we overflow again

2^2	2^1	2^0
1	0	0

4 = 0b100
0b11 + 0b1 = 0b100 just like 99 +1 = 100 in base 10

Now we can add anything!

If we remember the overflowing, adding bigger numbers
is just the same.

100110 +
010101

111011